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Key Points

In this chapter we will consider computations involving *matrices*. Here is a preview of the central results:

- **Algorithmic structure** will play a prominent role. Since linear algebra methods are part of the bedrock of computation on which other methods are built, computational efficiency is particularly important. Smart book-keeping is a large part of good methods. Considerations on computational complexity will significantly constrain the choice of probabilistic models, to a class of Gaussian distributions with specific factorisation structure.

- Do not be confused by the term *linear* algebra. Matrix inversion is in fact a **nonlinear** operation. This will force us to make a conscious decision when designing the probabilistic model underlying an algorithm. The model can either fit neatly with the likelihood and thus allow for more complicated observation (computation) paradigms, or fit cleanly with the latent quantity (the solution of a linear problem), and thus provide more explicit notions of posterior uncertainty. This is in contrast to integration, where the latent quantity (the integral) is linearly related to the observables (the integrand), so that both can be captured jointly in a Gaussian model.

- Certain properties of a matrix (for example positive definiteness), even when known a priori, can be difficult to capture in a prior that still allows for computationally efficient inference. We may decide to *not* encode certain kinds of knowledge in favour of computational complexity. While this means that such knowledge can then not be used in the action rule of the

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algorithm, one may still require that resulting point estimates, and even error estimates, are consistent with it.

- Putting these abstract points together, we will arrive at a concrete class of prior distributions that, when combined with certain "natural" algorithmic choices, give rise to classic iterative solvers; in particular to the seminal method of conjugate gradients. In contrast to the integration chapter, however, we will find that **calibrating uncertainty** is a much more intricate and challenging issue in linear algebra. The reason for this, put simply, is that matrices are big objects: their quadratic size means that linearly many matrix-vector projections as observations identify only a small part of the matrix.

- Each chapter has a wider conceptual point that transcends the concrete numerical setting. In the case of this chapter, it is the observation that the design of a good computational prior is a trade-off between constraints set by knowledge and those set by computational considerations. Even though one may patently know some things to be true, it may be counter-productive to include this information in the prior when doing so makes the computation considerably more complex.